Non-Graded Optional Project: Step-by-Step Instructions to the Learner

Project Title:
Building a GenAI-Powered Algo Trading Strategy

Objective

The objective of this project is to apply the concepts of Generative AI and algo trading covered in the course to create a functional trading strategy. This includes analyzing financial data, integrating real-time sentiment analysis, implementing machine learning techniques, and automating trading decisions. The project emphasizes both technical skills and financial insights to build a robust, practical trading solution.

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Project Description

In this project, students will create a Python-based trading system that combines technical indicators and sentiment analysis to generate actionable trading signals. The system will:

1. Fetch historical and real-time stock price data using APIs like Yahoo Finance. Your role involves:
2. Implement and optimize technical trading strategies such as moving average crossovers.

3. Incorporate Generative AI models for sentiment analysis of financial news and social media data.
4. Generate buy/sell signals by combining technical and sentiment indicators.
5. Visualize financial data, trading signals, and portfolio performance.
6. Provide detailed insights into the risk-reward dynamics of the trading strategy.

Steps of the Project

Step 1: Setting Up the Environment

1. Install the required Python libraries: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Yfinance, Transformers, Tweepy, and Requests.
2. Set up an IDE like PyCharm, IntelliJ IDEA, or Jupyter Notebook.
3. Obtain necessary API keys for financial data (Yahoo Finance or Alpha Vantage), news sentiment analysis (NewsAPI), and Twitter sentiment analysis (Twitter Developer Account).

Step 2: Data Collection

1. Historical Price Data:

- Use Yahoo Finance or Alpha Vantage API to fetch historical stock data (e.g., Tesla, Apple, or a user-selected stock) for the last 5 years.
- Ensure data includes open, high, low, close, and volume fields.

2. Real-Time Data:

- Set up a script to fetch real-time stock prices and sentiment in 1-minute intervals.

3. Sentiment Data:

- Use NewsAPI to fetch recent financial news headlines.
- Use Tweepy to collect recent tweets mentioning the selected stock.

Step 3: Data Preprocessing and Visualization

1. Clean and preprocess the collected data for analysis.
2. Create visualizations:
   - Line charts for historical stock price and moving averages.
   - Heatmaps for sentiment distribution.
   - Bar charts for portfolio performance.

Step 4: Sentiment Analysis

1. Use Hugging Face Transformers to analyze sentiment of news headlines and tweets.
2. Classify sentiment as Positive, Negative, or Neutral with confidence scores.
3. Generate a "Sentiment Score" for each day by combining scores from news and tweets.

Step 5: Technical Analysis

1. Implement a Moving Average Crossover Strategy:
    - Calculate 5-day and 20-day moving averages.
    - Generate buy/sell signals based on crossover events.
2. Optimize moving average parameters using a grid search for maximum profitability.

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Step 6: Strategy Integration

1. Combine technical and sentiment indicators to refine trading signals:
   - Buy when sentiment is positive, and the moving average crossover indicates an uptrend.
   - Sell when sentiment is negative, and the moving average crossover indicates a downtrend.
2. Automate decision-making to generate actionable trading signals.

Step 7: Machine Learning Enhancements

1. Use LSTM (Long Short-Term Memory) networks to predict future stock price trends based on historical data.
2. Visualize predicted prices alongside actual prices.
3. Adjust trading decisions based on LSTM predictions.

Step 8: Portfolio Simulation and Evaluation

1. Simulate a portfolio with an initial capital of $10,000.
2. Track cumulative returns, daily returns, and Sharpe Ratio.
3. Evaluate the risk-reward performance of the strategy.
4. Visualize portfolio growth over time.

Step 9: Reporting and Presentation

1. Create a detailed project report including:
   - Objectives, methodology, results, and insights.
   - Key visualizations and their interpretation.
   - Challenges faced and solutions implemented.
2. Submit a Python script file with comments explaining each section.

Instructions for Documenting and Sharing the Project for Peer Review

1. Document the Project

- Use word processing software to create your market analysis report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

2. Share the Project

- Save your report and presentation in PDF format.
- Upload your documents to the “Peer Review” area in the course shell.
- Provide a brief description of your project in the submission post.

3. Peer Review

- Review the projects submitted by your peers.
- Provide constructive feedback and comments on their work.

By completing this project, you will gain valuable insights into market analysis and develop skills that are essential for business strategy and decision-making.